DATA PRODUCT PORTS

A data product is created with the end user in mind, and as we know, users have different needs and preferences. For example, a data scientist might prefer to have data about movies rented in Latin America in the form of a well-described file. A data engineer might like to have that same data as a bunch of files to be processed with a crafted Python script. A programmer might prefer to use a REST API.

A data product is not only a form in which data is stored but also the various ways that data can be accessed. In the case of a data product, we call these *ports*.

A data product can have one or more *output ports*, and the infrastructure should make it easy to expose the same data using different forms. Next, we present examples of output port representations.

DATABASE-LIKE STORAGE

One of the primary ports that will come into play when developing data products is database-like storage, i.e., SQL databases, NoSQL databases, or data warehouse data marts. In many cases, this may be the only type of port provided by a data product. As a result, it offers excellent opportunities, especially in the analytical processing of large amounts of data, including the big-data area.

This type of storage can be queried with SQL language or its analog, giving basically unlimited possibilities of further processing. But, of course, they are addressed mainly to technical users who can work with this type of storage and use SQL language.

FILES

The second type of port, which will be familiar to less technical users of data products, is files that can be imported into tools such as Excel, Google Sheets, or used by data scientists. It is worth remembering that data engineers often feel confident working with this type of data source. Files allow working with large amounts of data, and many algorithms